Tourism Pressure and Housing Markets in Barcelona: A Counterfactual Analysis of HUT License Removal

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Abstract

This study examines the impact of removing Housing Units for Tourists (HUTs) on Barcelona's housing market, focusing on rental and selling prices across neighborhoods. Using spatial datasets, socioeconomic indicators, and housing price records, we construct a counterfactual scenario in which all HUTs are eliminated. The methodology integrates neighborhood-level harmonization, K-means clustering, spatial visualization, and regression-based impact estimation. Results show a clear asymmetry: rental prices tend to increase after HUT removal, while selling prices tend to decrease, particularly in high-tourism areas. These effects reflect structural heterogeneity across neighborhoods and align with urban economic theory regarding tourism-driven housing dynamics. The findings contribute to evidence-based policymaking by quantifying the differentiated effects of tourism regulation on local housing markets.

1. Introduction

Barcelona has experienced sustained pressure from tourism, driven in part by the proliferation of Housing Units for Tourists (HUTs). These units have been associated with rising housing prices, neighborhood transformation, and tensions between tourism and residential needs. In response, the municipality announced the cancellation of all HUT licenses, raising questions about the potential effects on the housing market.

The objective of this study is to quantify the counterfactual impact of removing HUTs on rental and selling prices across Barcelona's neighborhoods. The research question is:

What would happen to housing prices if all HUTs were removed from the city?

This study is relevant for urban policy, housing affordability debates, and the design of tourism regulations.

2. Methodology

2.1 Data Sources

We integrated multiple datasets from Open Data BCN and municipal sources, including:

- HUT registry
- Hotel locations
- Rental and selling prices (2014–2025)
- Population and foreign population
- Per capita income

- Neighborhood shapefiles

All datasets were standardized using a unified neighborhood identifier (*neigh_norm*).

2.2 Data Harmonization

Neighborhood names were normalized through:

- Unicode cleaning
- Removal of accents and stopwords
- Dictionary-based corrections
- Cross-dataset consistency checks

2.3 Spatial Analysis

We produced:

- A map of HUT density
- A map of K-means clusters (k=3)
- Spatial overlays for interpretation

2.4 Clustering

K-means segmentation identified three neighborhood profiles:

1. High-income residential
2. Touristic and central
3. Urban mixed areas

2.5 Counterfactual Modeling

We estimated:

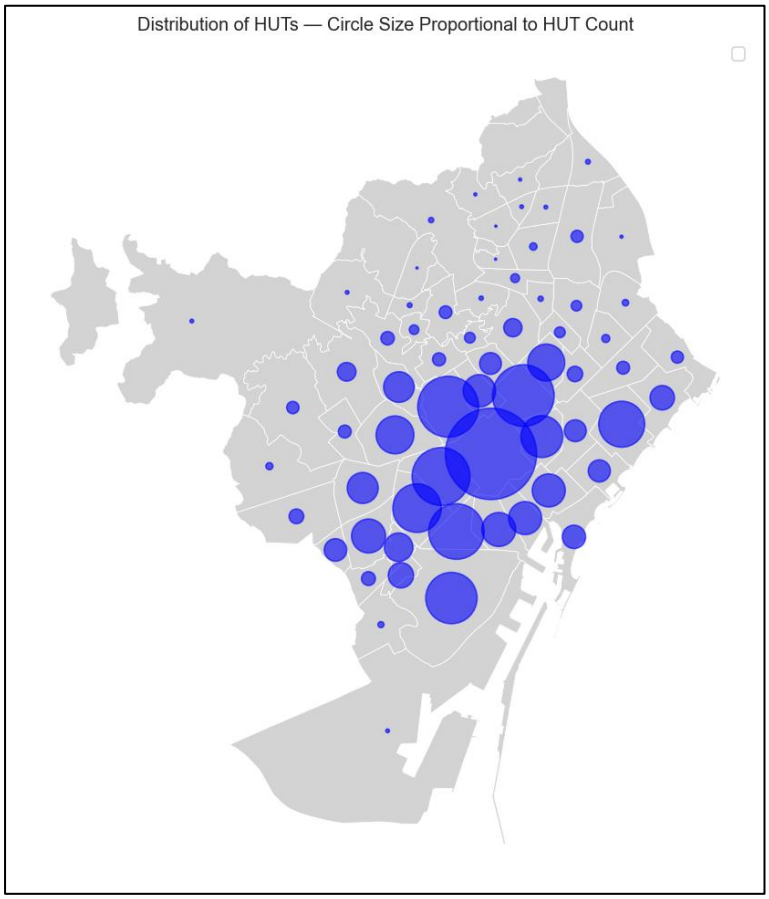
- Baseline predicted prices (with HUT density)
- Counterfactual predicted prices (HUT density = 0)
- Percentage impact for rental and selling markets

2.6 Statistical Analysis

We used:

- Linear regression
- LOESS smoothing
- Density estimation (KDE)

Figure 1. HUT Density Across Neighborhoods



3. Results

3.1 Price Impact Distribution

The KDE distribution shows:

- Rental price impacts cluster above zero (peak ≈ +2%)
- Selling price impacts cluster below zero (peak ≈ –2%)

3.2 Spatial Patterns

Touristic neighborhoods exhibit:

- Higher HUT density
- Stronger price impacts

3.3 Clustering

K-means segmentation reveals:

- Cluster 1 (touristic-central) aligns with the strongest impacts
- Cluster 0 (high-income residential) shows moderate effects
- Cluster 2 (urban mixed) shows smaller impacts

Figure 2. Distribution of Price Impacts After HUT Removal

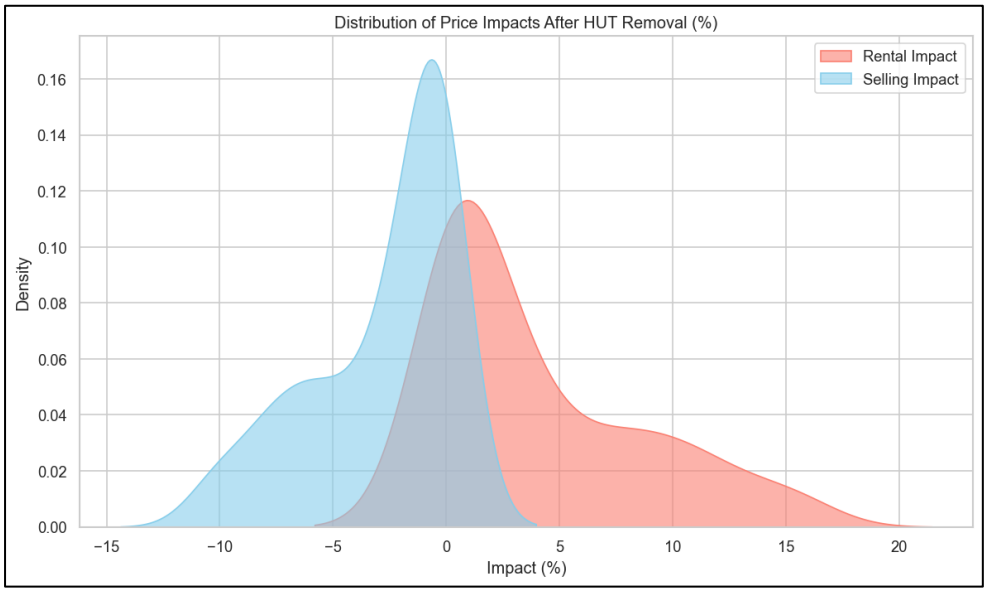
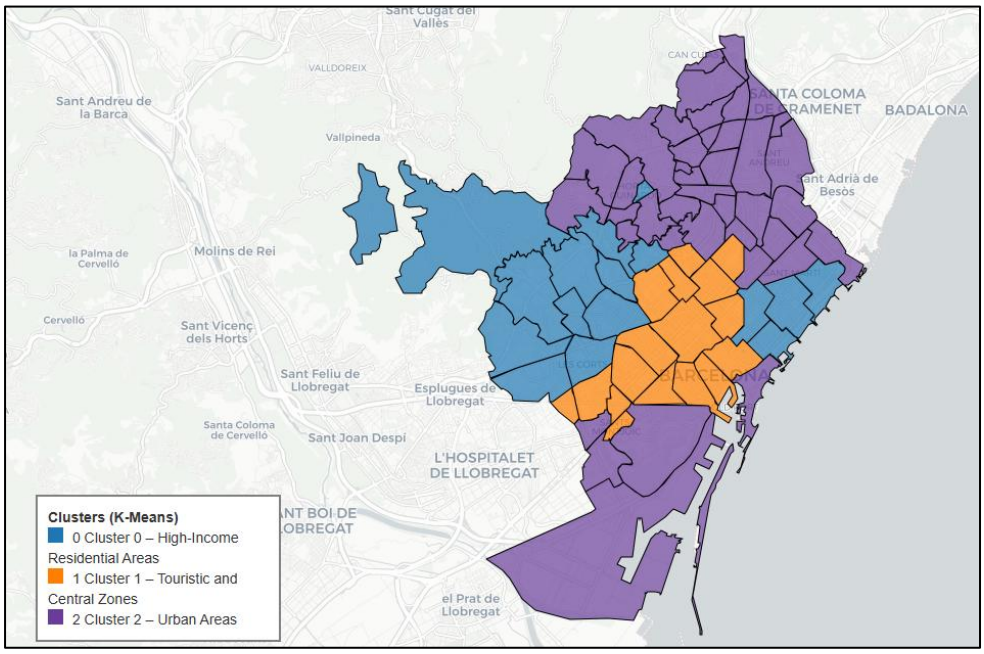


Figure 3. Neighborhood Clusters (K-means)



4. Discussion

The results indicate that removing HUTs produces **asymmetric effects** on the housing market. Rental prices increase because units previously used for tourism re-enter the long-term market, raising demand in central areas. Selling prices decrease due to reduced investment attractiveness and lower expected returns from tourist rentals.

These findings align with previous research on tourism-driven housing dynamics, which highlights the role of short-term rentals in inflating property values and reshaping urban

centers. The spatial heterogeneity observed reinforces the need for geographically targeted policies.

Limitations include:

- Annual data granularity
- Potential omitted variables
- Assumptions in the counterfactual model

Future research could incorporate dynamic modeling, resident surveys, or alternative regulatory scenarios.

5. Conclusion

This study provides empirical evidence that removing HUTs affects rental and selling prices differently across Barcelona's neighborhoods. The counterfactual analysis reveals that tourism regulation has significant and spatially uneven consequences.

References

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